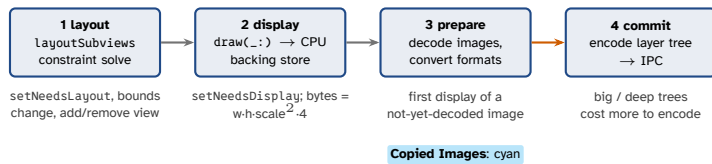
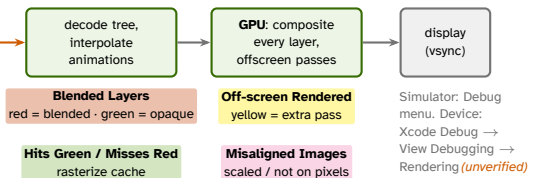


In one line: A CALayer is a **model object**: a bitmap (contents) plus geometry and animatable properties — no events, no layout constraints. Each run-loop turn your changes are **committed** (layout → display → prepare → commit) to the out-of-process **render server**, which interpolates animations and composites every layer on the GPU. Speed = a cheap commit + as few GPU passes as possible (no needless **blending**, **offscreen** passes, **copied** or **misaligned** images).

your process — main thread, one commit per run-loop turn



render server + GPU — a separate process (backboardd (unverified!))



once an animation is committed the server renders every frame itself — no further app work

How it works

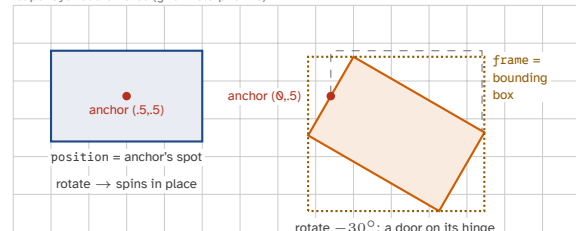
- Geometry:** bounds (own coordinates, size) · position (in the *superlayer*, where the anchorPoint sits) · anchorPoint (unit space, default (0.5, 0.5); every transform pivots here) · frame is *derived* and not directly animatable — animate bounds, position, transform. UIView.frame is **undefined** once transform is not identity: use center + bounds.
- Properties:** contents (a CGImage), contentsScale (1.0 on a standalone layer — set it to the display scale; a view sets its own), contentsGravity, contentsCenter (9-slice stretch), cornerRadius + cornerCurve = .continuous + maskedCorners, shadow*, transform (CATransform3D; m34 = -1/d for perspective), sublayerTransform, zPosition.
- Specialised layers** draw on the GPU, not in draw(·): CAShapeLayer (path; animate strokeEnd for a progress ring), CAGradientLayer, CAReplicatorLayer (instanceCount, instanceDelay: loading dots), CAEmitterLayer, CATextLayer, CATransformLayer (real 3D hierarchy), CAMetalLayer, AVPlayerLayer.
- Explicit animations:** CABasicAnimation(keyPath:·) (from/to/by), CAKeyframeAnimation(values + keyTimes, or a path), CASpringAnimation(mass, stiffness, damping, settlingDuration), CAAnimationGroup, CATransition(fade, push). add(·:forKey:·) *copies* the animation; the same key replaces the running one. Key paths: "transform.rotation.z", "position.x". They drive only the presentation — set the model value yourself.
- Timing** — CAMediaTiming, adopted by layers *and* animations: beginTime (layer time; CACurrentMediaTime() + 0.3 delays), duration, repeatCount, autoreverses, fillMode(.backwards shows fromValue during a delay), speed (**multiplies down the tree**), timeOffset. Pause: set speed to 0 and timeOffset to the layer's current time (QA1673); scrub by changing timeOffset.
- Blending:** the GPU mixes every non-opaque layer with what is beneath. An opaque background colour, alpha 1 and isOpaque let it skip that — e.g. give labels in cells a solid background.
- Rasterize cache** (WWDC14): about **2.5× the screen** in size; an entry unused for **100 ms** is evicted — so rasterizing things that come and go only re-renders them.
- CADisplayLink** fires once per frame: add to .main in .common mode (keeps firing while a scroll view tracks); it **retains its target** — invalidate() it; animate with targetTimestamp (the deadline), not timestamp; iOS 15 preferredFrameRateRange = CAFrameRateRange(minimum: maximum:preferred:) is a hint. On iPhone, above 60 Hz needs Info.plist CADisableMinimumFrameDurationOnPhone = YES; UIKit/SwiftUI/CAAnimation pace themselves.

Traps

- ! Standalone layer blurry on Retina — contentsScale is still 1.0.
- ! Setting anchorPoint and wondering why the view moved.
- ! Animating layer.frame, or reading view.frame after a rotation.
- ! CADisplayLink in .default mode stops during scrolling; never invalidated → its target (your VC) leaks.
- ! Implicit + explicit animation on one property fight — disable actions for the model set.
- ! isOpaque = true on a view that is not really opaque — undefined results.

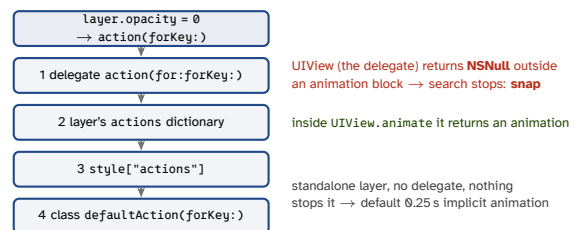
Geometry on a hinge

superlayer coordinates (grid = 0.5 pt units)



Changing anchorPoint keeps position fixed, so the layer **jumps**: compensate position by the same offset (× bounds.size).

The action search



Example — explicit, pause, resume

```
CATransaction.begin(); CATransaction.setDisableActions(true)
ring.strokeEnd = 1 // MODEL = final value, no implicit anim
CATransaction.commit()
let a = CABasicAnimation(keyPath: "strokeEnd")
a.fromValue = 0; a.duration = 0.6
a.timingFunction = CAMediaTimingFunction(name: .easeOut)
ring.add(a, forKey: "progress") // presentation 0 → 1, then stays at 1
func pause(_ l: CALayer) { // Apple QA1673
    let t = l.convertTime(CACurrentMediaTime(), from: nil)
    l.speed = 0; l.timeOffset = t
}
func resume(_ l: CALayer) {
    let paused = l.timeOffset; l.speed = 1; l.timeOffset = 0; l.beginTime = 0
    l.beginTime = l.convertTime(CACurrentMediaTime(), from: nil) - paused
}
```

Remember

Layout · display · prepare · commit — then the server paints. Pivot = anchor; frame = derived; NSNull = no animation; speed 0 = pause.